# Gabriel Cárcamo

## Senior Backend Engineer — Python & AI

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### SUMMARY

More than 10 years as an engineer at Experian (2014–2025). From 2018 I developed credit-risk models at Experian and refined them for its banking clients (Chase, Citi, Bank of America) to run in production on mainframe, AIX/Linux and AWS: validating them, moving the data they ran on and producing their distribution reports, for models that processed 300M+ data points. Since late 2025 I have built LLM features on my own projects, and since April 2026 FastAPI and PostgreSQL backends with typed LLM outputs (Gastify, Gustify), plus a hackathon triage agent with deterministic guardrails and Langfuse tracing. I also built a private set of 23 Claude Code agents, in regular use since January 2026. I learn the stack the work calls for: at Experian I started with Shell, SQL and the mainframe and added C, Python and AWS along the way, and in my own projects I moved from Flask to FastAPI in late 2025.

### BACKEND ENGINEERING · side projects, since April 2026

- **APIs & services:** FastAPI (async endpoints, SQLAlchemy 2.0 async, Alembic) and PostgreSQL 16, with REST API design, built for two product rebuilds, Gastify and Gustify (April–July 2026).
- **Testing & CI:** GitHub Actions with path-filtered pipelines; in Gastify, CI fails if the data-isolation tests (Postgres row-level security) against real Postgres 16 were skipped.

### AI ENGINEERING — LLMS & AGENTS · since late 2025

- **Gastify** (expense tracker for the Chilean market; side project, rebuild in progress, test deployment): Gemini output typed through PydanticAI, then an arithmetic check (items + tax − discount = total) that saves the receipt with a review flag when the totals do not add up; a plain-code PDF parser for known statement layouts calls AI only as a consented fallback.
- **SRE Triage Agent** (AgentX Hackathon, built solo with Claude Code, April 7–9, 2026; runs locally): FastAPI + async SQLAlchemy + PostgreSQL; deterministic guardrails before any model call; one structured LLM call per triage, with curated notes on the Solidus e-commerce codebase, through Claude Haiku 4.5 (Anthropic SDK, the default) or Gemini 2.5 Flash (LangChain, optional), plus an experimental engine on the Anthropic Managed Agents API (beta) that explores the repo; every file path the model cites is checked to exist in the repo; one trace per triage in a self-hosted Langfuse, via its SDK.
- **Gustify** (cooking app that suggests recipes from your pantry; side project, in development): FastAPI and PostgreSQL backend; a Gemini recipe generator with schema-typed outputs (google-genai with Pydantic response schemas) is built but switched off in the deployed app, which serves a curated catalog.
- **Agent development:** a private set of 23 Claude Code agents (as of September 2026), in regular use since January 2026 for my own paperwork, research and side projects (tax questions, legal correspondence, audit documentation). Each agent is a YAML definition plus a sidecar of method, knowledge and state, running cases through its own workflow; in 16 of the 23, escalation rules, written as prose instructions, tell the agent to stop and hand the decision to me or refer it to a professional. Deliverables are written for my review, and nothing is published or sent without my confirmation. Internal use; not client work.
- **Retrieval / RAG:** file-based context loading across the agent set, over the Markdown files in its sidecars (selective knowledge loading, pointer-based references to a single source of truth, context compression; no embeddings, vector store or reranking); familiar with the RAG spectrum (naive to agentic, Self-RAG, CRAG, Graph RAG) and its trade-offs; learning FAISS for vector search. *No vector-RAG deployment yet; file-based retrieval to date.*
- **AI-assisted development:** Claude Code is my main coding tool (custom hooks, skills, multi-instance orchestration), inside a test-first, review-gated workflow; earlier, GitHub Copilot at Experian (2024) for model implementation; MCP servers and tools inside Claude Code.

- **Gabe Suite** (self-built Claude Code plugin, since February 2026): 28 skills as of September 2026, including a test-first loop (plan, failing test, code, review, commit, push) with hooks that enforce the order, plus a command center that generates a project's documentation (entities, endpoints, tests, coverage, open work) from git, the test run and the plan files.

## INDEPENDENT PROJECTS · 2025 – now

Remote, Chile. Side projects listed above: FastAPI and PostgreSQL backends (Gastify, Gustify) since April 2026, with typed LLM outputs in Gastify, and a solo build for the AgentX Hackathon.

## EXPERIENCE — EXPERIAN · 2013–2025

**Senior Software Development Engineer** 2020 – Apr 2025

Experian · Remote, Chile · HR employment certificate (PDF)

- Deployed and hardened credit-risk models for Experian's banking clients (Chase, Citi, Bank of America) on mainframe, AIX/Linux and AWS. The focus was edge cases, test coverage and regulatory requirements, not tuning the models.
- Compiled the models in C and ran them as user-defined functions (UDFs) inside DB2, on mainframe and AIX/Linux: SQL queries ingested and joined the latest data, and I delivered the output to downstream platforms parsed into fields or as a single string with its layout sent separately, generated the data dictionaries and tested each build on the machines it would run on.
- Moved and validated data across platforms for models that processed 300M+ data points, with Python scripts I wrote that compare model outputs from mainframe, AIX/Linux and AWS runs.
- Built the Property Attributes feature engine: plans from requested attributes back to raw inputs, stops on missing data, loads only what the plan needs. Columnar pandas/NumPy and bitwise flags: from 4+ hours to 15 minutes.
- Designed, end to end, how data reached the models (the queries themselves) and how they worked inside: how they assembled and concatenated the attributes to compute. I wrote all of those functions.
- Fixed production issues for Experian's banking clients within 24–48 hours. As lead developer of Property Attributes, coordinated 4+ engineering teams; developers and data scientists used its engine via an internal portal to build and deploy models. Trained client and internal engineers on implementing models.

**Software Developer I & II** 2014 – 2020

Experian · Remote, Chile

- Modified mainframe JCL for notification-campaign programs that ran over 300M+ data points.
- Worked with DB2, MySQL and PostgreSQL: queries, creating and deploying tables, and index reviews on DB2.
- Supported critical systems 24/7 for internal and external teams, and built internal tools: Shell libraries, file-format converters and, from 2018, small Python CLI tools.

## SKILLS

**Languages:** Python (about 4 years professional), SQL (10+ years — DB2, PostgreSQL, MySQL), Shell/Bash (10+ years), C/C++, JavaScript/TypeScript

**Backend (side projects, 2026):** FastAPI, SQLAlchemy 2.0 (async), Alembic, PostgreSQL 16, RESTful API design; earlier Flask apps

**Platforms (Experian):** IBM mainframe (JCL, DB2), AIX/Linux, AWS · Scrum/Kanban, backup Scrum Master

**AI / LLM:** Claude API, Gemini API, PydanticAI, AI agents, prompt engineering, MCP, Langfuse tracing (self-hosted, SDK), OpenTelemetry (console spans), context engineering; RAG (conceptual; learning FAISS)

**Cloud / DevOps:** AWS (Lambda, EC2, S3; cloud migration work at Experian), Docker, GitHub Actions CI/CD, Git

**Data:** moving and validating model data across mainframe, AIX/Linux and AWS, pandas / numpy

## EDUCATION & LANGUAGES

**Ingeniero en Informática** (equivalent to a B.S. in Computer Science), Universidad Mayor, Chile, 2016–2020

**Spanish:** native · **English:** IELTS Academic 8.0 (2020), CEFR C1; daily working language